Week 3 Report: DINOv2 Representations & Attention on DESI

1. Key Concepts

- **DINOv2:** An improved vision transformer that uses Masked Image Modeling (MIM) and better regularization to learn highly effective patch-level features without supervision.
- **Attention Maps:** Visualizations of where the model's CLS token focuses its attention weights. They naturally outline objects in the image without requiring segmentation labels.
- **Registers:** Extra learnable tokens added to the model to absorb useless background information. This prevents the bright "artifact spikes" that otherwise appear when the model tries to store global data in random background patches.
- **Probing:** Freezing the pretrained model and training a simple classifier (Linear or MLP) on its extracted features. This tests how well the frozen layers already understand concepts like galaxy morphology.

2. Results & Comparison with the Paper

Our probing experiments evaluated the DINOv2 ViT-S/14 features on the 10-class Galaxy Zoo (gz10) dataset.

- **Our Results:** The MLP probe achieved **66.2%** accuracy on the Last Layer CLS (L12), and peaked at **~67.9%** when concatenating the CLS tokens from the last four layers (L9-L12).
- **Comparison:** This is highly consistent with the AstroDINO paper, which reported exactly **67%** accuracy for the DINOv2 ViT-S/14 model on this same task.

3. Answers to Required Questions

1. Does attention isolate the main galaxy, or every bright source in the cutout? It tends to highlight every bright source. The attention maps capture the central galaxy but also prominently highlight companion galaxies and bright foreground stars.

2. Does naive thresholding produce usable galaxy masks? Where does it fail? It produces a rough, basic outline but is not usable for precise segmentation. It fails by completely ignoring faint outer structures (like spiral arms or extended halos) and improperly blending nearby stars or artifacts into the main mask.

3. Which layer (or combination) represents galaxy morphology best? The concatenated CLS features from the last 4 layers (L9-L12) performed the best (~67.9% accuracy). For

individual layers, the final layer (L12) was the strongest, confirming that deeper layers capture the necessary global semantic information. Mean-pooling patch tokens performed worse than the CLS tokens.

4. Which classes get confused, and why? The model primarily confuses visually similar sub-classes. For example, different variations of spiral galaxies (e.g., tight vs. loose spirals) or smooth galaxies with slightly different ellipticities overlap heavily because their structural differences are very subtle at a 14x14 patch resolution.

5. How well do natural-image (ImageNet-style) features transfer to astronomical images? They transfer surprisingly well. Achieving ~68% accuracy on a complex 10-class scientific morphology task—using frozen features trained entirely on everyday Earth photos—demonstrates that DINOv2 learns highly generalizable shape and structure representations.